

OSSP:

Commands:

Uname : It displays the System information

Lscpu : It displays the complete CPU information

```
root@LAPTOP-4H877K4U:~# nano process.c
root@LAPTOP-4H877K4U:~# gcc process.c -o process
root@LAPTOP-4H877K4U:~# ./process
Enter Linux command: lscpu

Child Process
Child PID = 835
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:         39 bits physical, 48 bits virtual
Byte Order:            Little Endian
CPU(s):                16
On-line CPU(s) list:   0-15
Vendor ID:              GenuineIntel
Model name:             13th Gen Intel(R) Core(TM) i7-13620H
CPU family:             6
Model:                 186
Thread(s) per core:     2
Core(s) per socket:     8
Socket(s):              1
Stepping:               2
```

Ps : It displays the running process of our file

```
root@LAPTOP-4H877K4U:~# nano process.c
root@LAPTOP-4H877K4U:~# gcc process.c -o process
root@LAPTOP-4H877K4U:~# ./process
Enter Linux command: ps

Child Process
Child PID = 910
  PID TTY          TIME CMD
  755 pts/2        00:00:00 bash
  842 pts/2        00:00:00 process
  910 pts/2        00:00:00 ps

Parent Process
Parent PID = 842
Child PID = 910
```

Relation Between Hardware Resources and Operating System Services:

1. CPU

The operating system shares the CPU among multiple programs by giving each one a turn to execute.

Example: When we open Chrome and a music player together, the OS schedules CPU time for both. It Process Both Chrome And music player

2. Memory (RAM) -> Memory Management

The operating system gives memory to each program that is running
It makes sure one program does not use another program's memory.
Example: Chrome and VS Code each get their own memory space while running.

3. Storage (SSD) → File System Management

The operating system stores and organizes files on the storage device.
Example: Saving a document in the Documents folder on our computer.

4. I/O Devices → Device Management

The operating system uses device drivers to control input and output devices.